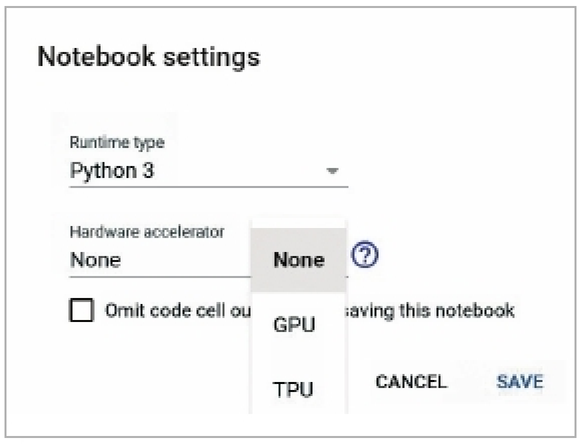

Figure 7: Activation of the hardware accelerator with Google Colaboratory notebook

Google Colaboratory is available as a Google app in Google Cloud Services. It can be invoked from Google Drive as depicted in Figure 6 or directly at *https://colab.research.google.com*.

The Jupyter notebook in Google Colaboratory is associated with the CPU, by default. If a hardware accelerator is required, like the tensor processing unit (TPU) or the graphics processing unit (GPU), it can be activated from *Notebook Settings*, as shown in Figure 7.

Figure 8 presents a view of Python code that is imported in the Jupyter Notebook. The data set can be placed in Google Drive. The data set under analysis is mapped with the code so that the script can directly perform the operations as programmed in the code. The outputs and logs are presented on the Jupyter Notebook in the platform of Google Colaboratory.

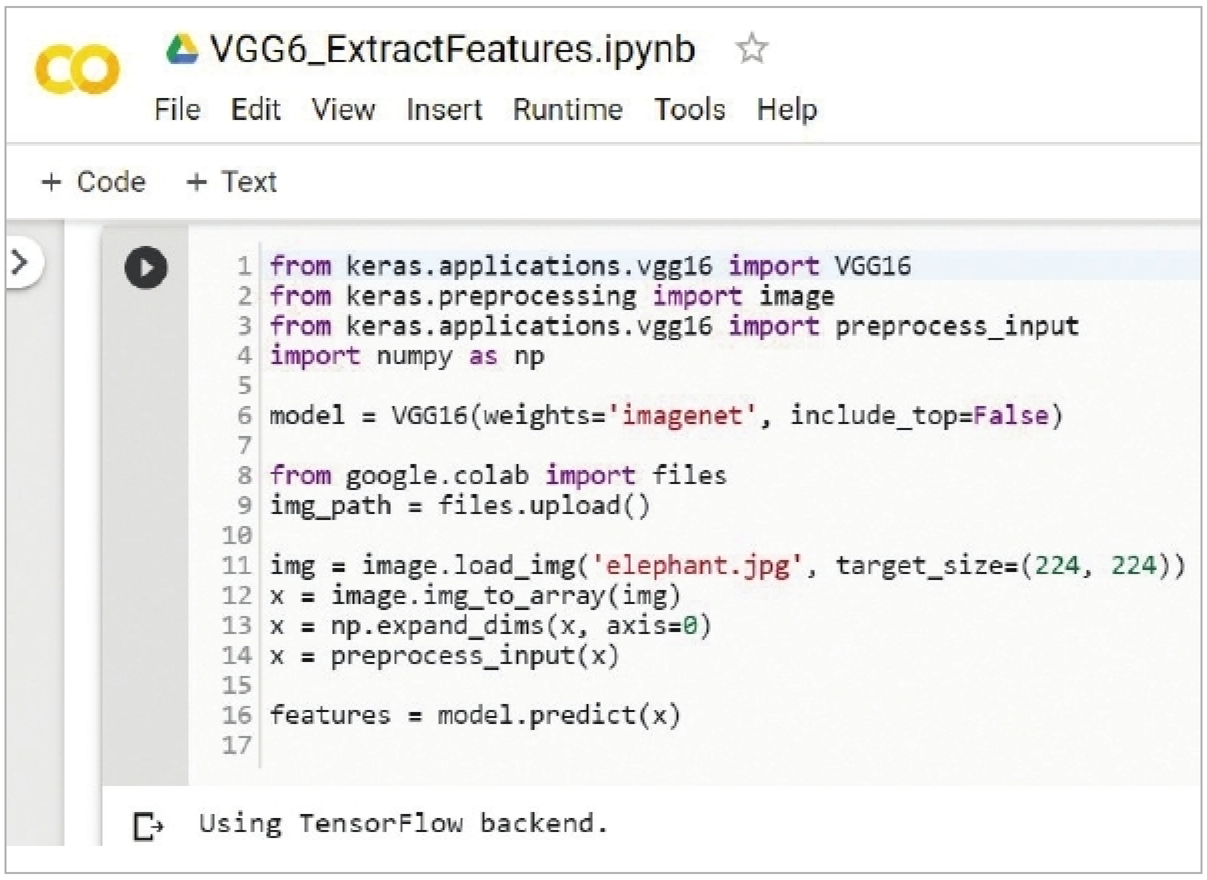

Figure 8: Implementation of the Python code on the Google Colaboratory Jupyter Notebook

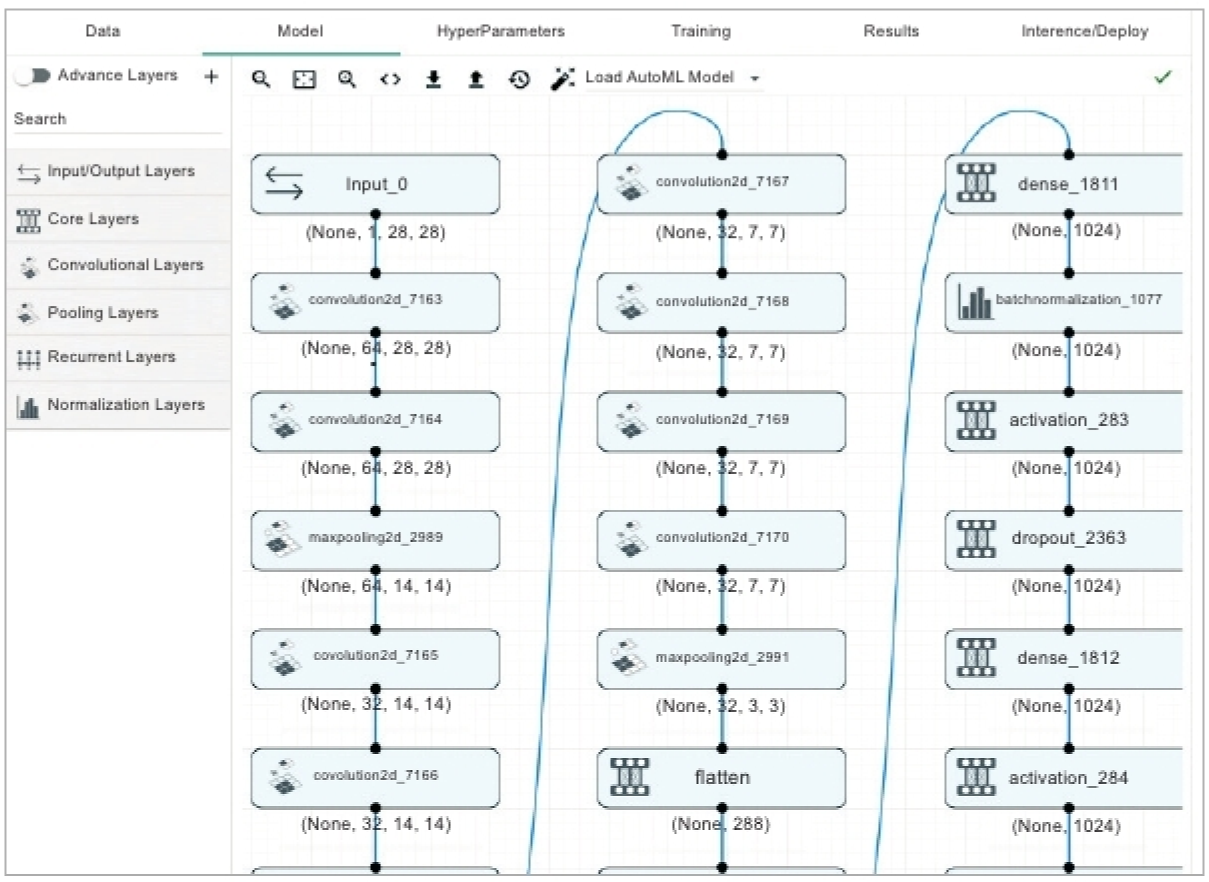

Figure 9: Importing layers in neural network models on Deep Cognition

Deep Cognition

URL: *https://deepcognition.ai/*

Deep Cognition provides the platform to implement advanced neural networks and deep learning models. AutoML with Deep Cognition provides an autonomous integrated development environment (IDE) so that the coding, testing and debugging of advanced models can be done.

It has a visual editor so that the multiple layers of different types can be programmed. The layers that can be imported are core layers, hidden layers, convolutional layers, recurrent layers, pooling layers and many others.

The platform provides the features to work with advanced frameworks and libraries of MXNet and TensorFlow for scientific computations and deep neural networks.

Scope for research and development

Research scholars, academicians and practitioners can work on advanced algorithms and their implementations using cloud based platforms dedicated to high performance computing. With this type of implementation, there is no need to purchase the specific infrastructure or devices; rather, the supercomputing environment can be hired on the cloud. END

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